Problem sheet 3 for Rigid Analytic Geometry Summer Term 2026

Let K be a field with an absolute value $|\cdot|_K$ satisfying the conditions from sheet 1. Let L/K be a finite field extension.

Problem 1 (2 points)**.** *There is a norm $\|\cdot\,|L\|$ on the K-vector space L such that $\|xy\,|L\| \leq \|x\,|L\|\,\|y\,|L\|$ for all $x, y \in L$.*

Problem 2 (4 points)**.** *For every norm $\|\cdot\,|L\|$ on the K-vector space L,*

$$|l|_L = \lim_{n\to\infty} \sqrt[n]{\|l^n\|}$$

is a well-defined and does not depend on the choice of $\|\cdot\,|L\|$.

It is clear from this that if any extension of $|\cdot|_K$ to an ultrametric absolute value on L exists, it must equal $|\cdot|_L$. We would like to show that $|\cdot|_L$ is indeed an ultrametric absolute value on L extending $|\cdot|_K$.

Problem 3 (3 points)**.** *We have $|xy|_L \leq |x|_L\,|y|_L$, and $|\cdot|_L$ is a norm on the K-vector space L.*

Problem 4 (1 point)**.** *Show that $|k|_L = |k|_K$ when $k \in K$!*

Problem 5 (2 points)**.** *For every $\ell \in L$, there is a norm $\|\cdot\,|L\|$ on the K-vector space L such that $\|\ell x\,|L\| = \|\ell\,|L\|\,\|x\,|L\|$.*

Problem 6 (1 point)**.** *Show that $|\cdot|_L$ is indeed an extension of $|\cdot|_K$ to an ultrametric absolute value on L!*

It is straightforward to see that L is complete with respect to this absolute value. Also, $|\cdot|_K$ hat a unique extension to any algebraic extension L/K, but L will only be complete if the extension is finite.

For a topological space X, recall that $\mathfrak{Qc}(X)$ is the set of quasi-compact open subsets of X. For every point x of a spectral space X, let S_x be the set of all $y \in X$ such that x is a specialization of y.

Problem 7 (2 points)**.** *Show that $S_x \cap S_y$ is the intersection of the $\Omega \cap \Theta$, taken over all $\Omega, \Theta \in \mathfrak{Qc}(X)$ such that $x \in \Omega$ and $y \in \Theta$!*

Problem 8 (2 points)**.** *Show that two points x and y of a spectral space X have disjoint neighbourhoods if and only if $S_x \cap S_y = \emptyset$ (in other words, if and only if they have no common generification).*

Problem 9 (4 points)**.** *Let $A \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation and $H \subseteq \Gamma$ a convex subgroup containing all $v(a)$ which are ≥ 1. Show that*

$$w(a) = \begin{cases} v(a) & v(a) \in H \\ 0 & \text{otherwise.} \end{cases}$$

defines a valuation of A!

One of the 21 points from this sheet are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday May 11.